
Supplement: The Price of Fidelity in Multiclass Online Recalibration

1 Explicit Simplex Net

Fix $h \in (0, 1)$ and put $m = \lceil \sqrt{K}/h \rceil$. Define

$$N_h = \left\{ z/m : z \in \mathbb{Z}_+^K, \sum_{i=1}^K z_i = m \right\}.$$

For $p \in \Delta_K$, first set $z_i = \lfloor mp_i \rfloor$. The deficit $m - \sum_i z_i$ is an integer between zero and $K - 1$. Add one to that many coordinates with largest fractional parts. The resulting vector still sums to m , and every coordinate differs from mp_i by less than one. Thus the map $R_h(p) = z/m$ satisfies

$$\|R_h(p) - p\|_2 < \frac{\sqrt{K}}{m} \leq h.$$

Stars and bars gives

$$|N_h| = \binom{m+K-1}{K-1} \leq C_K m^{K-1} \leq C_K h^{-(K-1)}.$$

This proves all net properties used in Lemma 1 of the main paper.

2 Strongly Proper Losses

A multiclass loss ℓ is λ -strongly proper in Euclidean norm if for every $p, q \in \Delta_K$,

$$\mathbb{E}_{Y \sim q}[\ell(p, Y) - \ell(q, Y)] \geq \lambda \|p - q\|_2^2.$$

The Brier loss in the main paper has equality with $\lambda = 1$.

Theorem 1 (Strongly proper extension). *Fix $K \geq 2$ and $\lambda > 0$. There are constants $c_{K,\lambda}, C_{K,\lambda}, \beta_{K,\lambda} > 0$ such that, on the truthful product instance from the main paper, every learner with*

$$\mathbb{E} \left[\frac{1}{T} \sum_{t=1}^T (\ell(p_t, y_t) - \ell(q_t, y_t)) \right] \leq \beta$$

for a λ -strongly proper loss obeys

$$\mathbb{E} \text{Cal}_1(p, y) \geq c_{K,\lambda} \min \left\{ 1, (T\beta^{(K-1)/2})^{-1/2} \right\} - C_{K,\lambda} \sqrt{\beta}.$$

Proof. Conditional strong propriety and truthfulness give

$$\mathbb{E} \sum_{t=1}^T \|p_t - q_t\|_2^2 \leq \frac{\beta T}{\lambda}.$$

Choose $h = c\sqrt{\beta/\lambda}$. The rounding argument then gives $\mathbb{E}B \leq C_K h^2 T$. Every remaining lemma depends only on this movement bound, the source lattice, and the interior label variance. Therefore the proof in the main paper gives

$$\mathbb{E} \text{Cal}_1(p, y) \geq c_K \min \{1, (Th^{K-1})^{-1/2}\} - C_K h.$$

Substitution absorbs powers of λ into the displayed constants. $\square$

The theorem requires only a lower curvature bound. It does not require smoothness, mixability, or bounded loss values beyond integrability of the stated expected excess loss.

3 Intrinsic Hint Dimension

We prove Proposition 1 of the main paper. Fix $1 \leq r \leq K - 1$. Choose an r -dimensional affine box Q_r inside the simplex by varying the first r coordinates in a sufficiently small positive interval, holding coordinates $r + 1$ through $K - 1$ at positive constants, and using the last coordinate as the remainder. A spacing- h grid in Q_r has $M = \Theta_{K,r}(h^{-r})$ points and the same interior variance bound used in the main proof.

Let P_{Q_r} be Euclidean projection onto the closed convex box Q_r , which has an h -net $N_{h,r}$ of size at most $C_{K,r}h^{-r}$. Define r_t by projecting p_t and rounding the projection to this net. Since every q_t lies in Q_r , the projection inequality gives

$$\|P_{Q_r}(p_t) - q_t\|_2 \leq \|p_t - q_t\|_2$$

and

$$\text{dist}(p_t, Q_r) \leq \|p_t - q_t\|_2.$$

The grouping argument from the main rounding lemma now gives

$$\text{Cal}_1(r, y) \leq \text{Cal}_1(p, y) + \frac{1}{T} \sum_{t=1}^T \text{dist}(p_t, Q_r) + h.$$

Under expected fidelity at most β , Jensen and Cauchy–Schwarz bound the expected projection term by $\sqrt{\beta}$. The rounded movement still obeys

$$\mathbb{E} \sum_{t=1}^T \|r_t - q_t\|_2^2 \leq C_{K,r}(\beta + h^2)T.$$

All rounded bins now lie in the r -dimensional affine space. Repeat the martingale proof with

$$s = 1 + \frac{2}{r}, \quad a = \frac{2s-1}{2s-2}, \quad b = \frac{1}{2s-2}.$$

The source ball count and occupancy transport argument use dimension r . The net cardinality is $O_{K,r}(h^{-r})$. Thus the global occupancy moment is $O_{K,r}(T^s h^2)$ in the long-horizon regime, and the same exponent cancellation gives

$$\mathbb{E} \text{Cal}_1(r, y) \geq c_{K,r} \min\{1, (Th^r)^{-1/2}\} - C_{K,r}h.$$

Returning from r_t to p_t costs at most $\sqrt{\beta} + h$ in expectation. Setting $h = c\sqrt{\beta}$ proves Proposition 1.

4 Fractional Martingale Moment

We expand the interpolation step in Lemma 3 of the main paper. Fix an output bin v and let $Z = \mu_v$. Put

$$s = 1 + \frac{2}{d}, \quad \theta = \frac{s-1}{2s-1}.$$

The interpolation identity

$$\frac{1}{2} = \frac{\theta}{1} + \frac{1-\theta}{2s}$$

and log-convexity of L_p norms give

$$\|Z\|_2 \leq \|Z\|_1^\theta \|Z\|_{2s}^{1-\theta}.$$

Rearranging and raising to the required powers yields

$$\mathbb{E}|Z| \geq \frac{(\mathbb{E}Z^2)^a}{(\mathbb{E}|Z|^{2s})^b}, \quad a = \frac{2s-1}{2s-2}, \quad b = \frac{1}{2s-2}.$$

The interior variance and Burkholder–Rosenthal bounds therefore imply

$$\mathbb{E}|\mu_v| \geq c_K \frac{(\mathbb{E}\nu_v)^a}{(\mathbb{E}\nu_v^s)^b}.$$

For completeness, the Burkholder–Rosenthal inequality used here states that for a discrete martingale M_t with differences D_t and every real $p \geq 2$,

$$\mathbb{E} \max_{t \leq T} |M_t|^p \leq C_p \left(\mathbb{E} \langle M \rangle_T^{p/2} + \mathbb{E} \max_{t \leq T} |D_t|^p \right).$$

This is the form quoted by Collina et al. (2026, Lemma 1), who attribute it to Burkholder (1973). In our application, $D_t = \mathbf{1}\{r_t = v\}(q_{t,1} - y_{t,1})$. Predictability gives

$$\langle M \rangle_T = \sum_t \mathbf{1}\{r_t = v\} q_{t,1} (1 - q_{t,1}) \leq \nu_v.$$

Also, $\max_t |D_t|^{2s} \leq \mathbf{1}\{\nu_v > 0\} \leq \nu_v^s$. Hence $\mathbb{E} |\mu_v|^{2s} \leq C_K \mathbb{E} \nu_v^s$ for the noninteger value $2s = 2 + 4/d$.

To combine bins, define

$$A_v = \mathbb{E} \nu_v, \quad B_v = \mathbb{E} \nu_v^s, \quad X_v = A_v^a / B_v^b.$$

Since $a - 1 = b$, Hölder with conjugate exponents a and $a/(a - 1)$ gives

$$\sum_v A_v = \sum_v X_v^{1/a} B_v^{b/a} \leq \left(\sum_v X_v \right)^{1/a} \left(\sum_v B_v \right)^{b/a}.$$

As $\sum_v A_v = T$, rearrangement gives

$$\sum_v \mathbb{E} |\mu_v| \geq c_K \frac{T^a}{(\sum_v \mathbb{E} \nu_v^s)^b}.$$

5 Detailed Lattice Transport Proof

Let the source chart be $q(x) = (x_1, \dots, x_d, 1 - \sum_j x_j)$. For any $x, x' \in \mathbb{R}^d$,

$$\|q(x) - q(x')\|_2^2 = \|x - x'\|_2^2 + \left(\sum_j (x_j - x'_j) \right)^2 \geq \|x - x'\|_2^2.$$

Therefore a Euclidean ball of radius R in the simplex intersects no more source points than a radius- R ball intersects in the spacing- h Cartesian lattice. A coordinate cube cover gives at most $C_K(1 + R/h)^d$ distinct points. Since each source point occurs at most L times, the occurrence count is at most

$$C_K L (1 + R/h)^d. \tag{1}$$

Fix an output v and sort the distances of assigned source occurrences as $D_1 \leq \dots \leq D_{\nu_v}$. The center v need not lie on the lattice and may lie on the simplex boundary. Equation (1) still applies. There are constants c_K, C_K such that for every $i > C_K L$,

$$D_i \geq c_K h (i/L)^{1/d}.$$

Consequently,

$$\begin{aligned} b_v &= \sum_{i=1}^{\nu_v} D_i^2 \geq c_K h^2 L^{-2/d} \sum_{i=C_K L+1}^{\nu_v} i^{2/d} \\ &\geq c_K h^2 L^{-2/d} \left(\nu_v^{1+2/d} - C_K L^{1+2/d} \right). \end{aligned}$$

Moving the baseline term to the other side proves the occupancy transport inequality. The proof uses only an upper bound on source multiplicity, so it also covers a horizon that stops partway through the final lattice cycle.

Summing the transport inequality gives

$$\sum_v \nu_v^s \leq C_K \left(N_{\text{occ}} L^s + L^{2/d} B/h^2 \right),$$

where N_{occ} is the number of occupied output bins. If $T < M$, then $L = 1$ and $N_{\text{occ}} \leq T$. If $T \geq M$, then $L \leq C_K T h^d$ and $N_{\text{occ}} \leq |N_h| \leq C_K h^{-d}$. These two cases give the global occupancy lemma after taking expectations.

6 Expected Upper-Bound Conversion

We detail Proposition 2 of the main paper. Theorem 17 of Chen et al. (2026) gives a pure-forecast algorithm with expected cumulative Brier regret $O(\eta^2 T)$ relative to any reference algorithm. It also gives, with probability at least $1 - \delta$,

$$K_T \leq \tilde{O}_K \left(\sqrt{T} + \eta^{-d} + \eta^2 T + \log(1/\delta) \right),$$

where $K_T = T \text{Cal}_2^2$ and the notation hides fixed- K and logarithmic factors. Choose the reference algorithm that outputs the revealed hint q_t . Choosing $\delta = T^{-2}$ and using the deterministic bound $K_T \leq 2T$ on the failure event gives

$$\mathbb{E} K_T \leq \tilde{O}_K(\sqrt{T} + \eta^{-d} + \eta^2 T).$$

Jensen and $\sqrt{x+y+z} \leq \sqrt{x} + \sqrt{y} + \sqrt{z}$ yield

$$\mathbb{E} \text{Cal}_2 \leq \sqrt{\mathbb{E} K_T / T} \leq \tilde{O}_K \left(T^{-1/4} + \frac{1}{\sqrt{T} \eta^d} + \eta \right).$$

For $d \geq 2$, set $\eta = \varepsilon$ and $T = \tilde{\Omega}_K(\varepsilon^{-(d+2)})$. Then

$$T^{-1/4} \leq \tilde{O}_K(\varepsilon^{(d+2)/4}) \leq \tilde{O}_K(\varepsilon),$$

and the other two terms are also $\tilde{O}_K(\varepsilon)$. This explains why the multiclass upper theorem covers $K \geq 3$. The binary case uses the sharper direct construction of Hu et al. (2026).

7 Deterministic Adversarial Realization

The lower construction randomizes labels, while the online guarantee in the main paper quantifies over deterministic sequences and averages only over the learner seed. Suppose an algorithm met both target guarantees for every deterministic label realization under the fixed hint sequence. Averaging each guarantee over the independent categorical labels would preserve both inequalities. The price-of-fidelity theorem rules this out below the stated horizon. Therefore at least one deterministic label realization violates at least one target guarantee. This is the standard averaging form of the probabilistic method and requires no adaptive adversary.

8 Calibration Norms

For weights $w_v = n_v/T$ that sum to one, Cauchy–Schwarz gives

$$\text{Cal}_1 = \sum_v w_v \|v - \rho_v\|_2 \leq \left(\sum_v w_v \|v - \rho_v\|_2^2 \right)^{1/2} = \text{Cal}_2.$$

Thus every lower bound on expected Cal_1 applies directly to expected Cal_2 . In the binary case, identifying a scalar probability p with $(p, 1 - p)$ changes both norms by only a factor $\sqrt{2}$. No rate exponent is affected.

References

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